Justin Liu

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Education

Vanderbilt University **Aug 2026 - May 2029**
B.S. in Computer Science *Nashville, TN*

Technical skills

Languages: Bash, Java, JavaScript, Python, SQL
Frameworks & Libraries: Express.js, NumPy, Pygame, React.js
Developer Tools: Blender, Docker, Git, Figma, Node.js, Stripe

Experience

Undergraduate Research Assistant **Aug 2026 – Present**
Vanderbilt University *Nashville, TN*

- Operated Bambu and nScrypt 3D printers while collecting IMU, acoustic, and camera sensor data
- Calibrated a fisheye camera by estimating intrinsic and distortion parameters to generate pixel mappings to correct lens distortion, producing an oblique view of the print surface
- Applied a perspective transformation to convert the oblique view into a top and front projection for 3D-print monitoring

Software Engineering Intern **Mar 2025 – July 2025**
TruthBridge AI *Remote*

- Built a React.js search interface with nine filters, live result counts, and college news displays to query an Airtable database of 5,000+ colleges.

Software Engineering Intern **Apr 2024 – May 2024**
NYCDOE Division of Instructional and Information Technology *NYC, NY*

- Reviewed the NYC Department of Education website and proposed navigation, layout, and accessibility improvements.
- Presented redesign recommendations to senior leadership, securing adoption of 20+ proposed features

Projects

GymCoder | *Docker, Express.js, Node.js, Stripe, Google OAuth, Pyodide* **Apr 2026**

- Built a full-stack coding platform supporting Python, JavaScript, and Java execution, custom test cases, and progress tracking
- Implemented authentication, subscription billing, and persistent user state using Node.js, Express.js, SQLite, Google OAuth, and Stripe

Billiards Shot Solver | *Blender, NumPy, Pygame* **Sep 2026**

- Built a physics-based billiards simulator modeling ball collisions and cushion rebounds
- Developed a numerical search algorithm to compute angle, power, and spin for user-selected shots
- Created dimensionally accurate 2D and 3D pool table visualizations using Blender and real-world measurements

Hackathons & competitions

AEGIS **Oct 17-18 2026**
HackTN26 *Nashville, TN*

- Co-developed a hospital surge simulator with configurable patient arrivals, staffing, bed capacity, severity, and medical supplies to test emergency response scenarios.
- Integrated an AI assistant to propose resource changes, with analysis dashboards, patient-flow replay, and scenario comparisons to identify bottlenecks and evaluate tradeoffs.

ICPC Mid-Central USA Regional Qualifier **Oct 2026**